

Cellular Biology and Energy Systems

Topic: Photosynthesis and Cellular Respiration

Light Reactions take place in the thylakoid membranes.

Chlorophyll absorbs solar photon energy to split water molecules.

Calvin Cycle uses ATP and NADPH to fix carbon dioxide into glucose.

Topic: Cellular Respiration & ATP Production

Glycolysis breaks down glucose into pyruvate in the cytoplasm.

Krebs Cycle produces electron carriers NADH and FADH₂.

Prerequisites: Basic Organic Chemistry, Enzymes.

Electron Transport Chain drives ATP synthase across inner membrane.